AI Use Disclosure Report

HiMCM 2018 Problem B — Team GoHiMCM

25 October 2026

1 Purpose

This report lists AI tools used while building AeroTherm-MPC. It follows COMAP guidance on transparent AI disclosure. Human teammates remain responsible for all submitted claims.

2 Tools and roles

- **Cursor (Composer agent).** Assisted Python modules for the 2R2C thermal network ODEs, prediction matrices, and SciPy model predictive control (MPC) solver. Also drafted LaTeX structure under human review.
- **Local Ollama (`qwen2.5:7b`).** Designed for multi-inhabitant thermal preference dialogue in `cognitive_thermostat.py`. Agents include EMS, senior, and parent personas. The locked dialogue file records a Mock fallback when Ollama was offline. The code path still prefers live Ollama when `localhost:11434` responds.
- **No cloud LLM for numeric invention.** KPI tables cite only local CSV/JSON under `results/`.

3 What AI did not do

- AI did not fabricate locked simulation numbers.
- AI did not replace human judgment on design rules.
- The Summary Sheet numeric cells were checked against locked files by teammates.
- The homeowner advisory letter was edited by teammates for public clarity.

4 Human verification checklist

1. Re-ran teaching Bang-Bang and MPC day scripts locally.
2. Confirmed `bang_vs_mpc_summary.json` matches Table 1 in the paper.
3. Confirmed `harness_benchmark_results.csv` matches the harness table.
4. Confirmed `agent_thermostat_dialogue.json` matches Section 6 claims.
5. Confirmed paper figures under `paper_figures/` are 300 DPI exports from those runs.

5 Prompt log pointer

Detailed Cursor interactions are appended in `prompts/cursor_log.md`. That log is the machine-readable companion to this disclosure.

6 Statement

All numerical control results in the paper come from local Python simulations. AI tools assisted code and drafting only. Final responsibility rests with Team GoHiMCM.